

Question 1 (15 marks) Commence each question on a SEPARATE page

a. Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \tan^2 x \, dx$. **2**

b. Find $\int_0^3 \frac{5}{x^2 - 6x + 18} \, dx$ **3**

c. i. Find the real numbers a , b and c such that **3**

$$\frac{4x - 6}{(x + 1)(2x^2 + 3)} \equiv \frac{a}{x + 1} + \frac{bx + c}{2x^2 + 3}$$

ii. Hence, or otherwise, find $\int \frac{4x - 6}{(x + 1)(2x^2 + 3)} \, dx$. **2**

d. Show that $\int_0^1 \frac{2y}{1 + 2y} \, dy = 1 - \frac{1}{2} \ln 3$. **2**

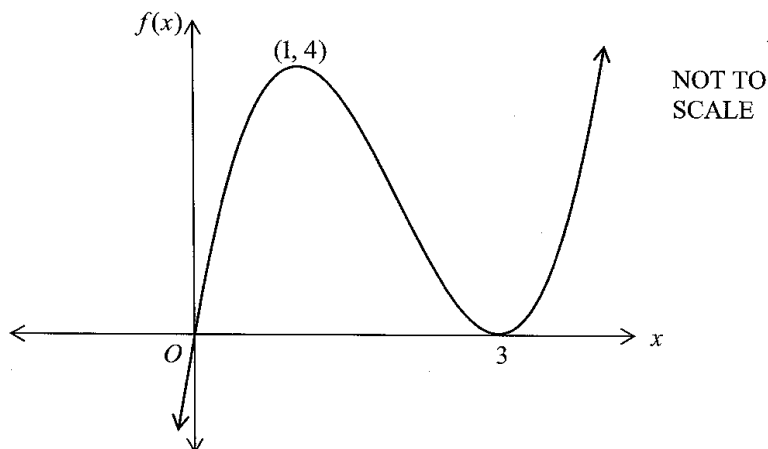
e. Using the substitution $t = \tan \frac{x}{2}$, evaluate $\int_0^{\frac{\pi}{3}} \frac{dx}{13 + 5 \sin x + 12 \cos x}$ **3**

Question 2 (15 marks) Commence each question on a SEPARATE page

- a. Simplify $(1 - 2i)^2$ **1**
- b. Solve $z^2 = 5 - 12i$, giving your answer in the form $x + iy$, where x and y are real. **2**
- c. i. Express $1 + i$ in modulus-argument form. **1**
- ii. Hence evaluate $(1 + i)^{12}$ **2**
- d. i. On an Argand diagram, shade the region where both $|z - 1| \leq 1$ and $0 \leq \arg z \leq \frac{\pi}{6}$. **2**
- ii. Find the perimeter of the shaded region. **2**
- e. The points O , A , Z and C on the Argand diagram represent the complex numbers 0 , 1 , z and $z + 1$ respectively, where $z = \cos \theta + i \sin \theta$ is any complex number of modulus 1 , with $0 < \theta < \pi$.
- i. Explain why $OACZ$ is a rhombus. **1**
- ii. Show that $\frac{z-1}{z+1}$ is purely imaginary. **2**
- iii. Find the modulus and argument of $z + 1$. **2**

Question 3 (15 marks) Commence each question on a SEPARATE page

- a. The function defined by $f(x) = x(x - 3)^2$ is drawn below.



Draw separate, one-third sketches, of the following:

- | | | |
|------|----------------------|----------|
| i. | $y = f(x)$ | 1 |
| ii. | $y = \frac{1}{f(x)}$ | 1 |
| iii. | $y^2 = f(x)$ | 2 |
| iv. | $y = \sin^{-1} f(x)$ | 2 |
| v. | $y = \log_e f(x)$ | 2 |

- b. Consider the curve given by the equation $x^2 - y^2 + xy + 5 = 0$.

- | | | |
|-----|--|----------|
| i. | Show that $\frac{dy}{dx} = \frac{2x+y}{2y-x}$. | 2 |
| ii. | Hence find the coordinates of the points on the curve where the tangent to the curve is parallel to the line $y = x$. | 2 |

- | | | |
|----|--|----------|
| c. | Show that the line $y = -3x$ is a tangent to the curve $f(x) = x + \frac{1}{x-1}$, and find the co-ordinates of the point of contact. | 3 |
|----|--|----------|

Question 4 (15 marks) Commence each question on a SEPARATE page

a. i. Given $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$. By writing $\cos^n x = \cos x \cdot \cos^{n-1} x$, **3**

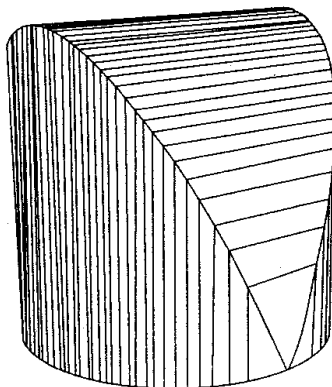
prove that $I_n = \frac{n-1}{n} \cdot I_{n-2}$, where $n = 2, 3, 4, \dots$

ii. Hence find the value of $\int_0^{\frac{\pi}{2}} \cos^4 x \, dx$. **3**

- b. For his visual arts major work, Lionel took a wooden cylinder and carved into it the shape shown in the diagram below.

The base of her shape is a circle with radius 8 cm.

Each vertical cross-section shown in the diagram is a square.



- i. Show that the area A of the cross-section distance x cm from the centre of the base is $A = 4(64 - x^2)$. **2**

- ii. Hence show that the volume V of the art project is given by **2**
- $$V = 8 \int_0^8 (64 - x^2) \, dx \text{ and evaluate the integral.}$$

- c. i. If α is a double root of the polynomial equation $P(x) = 0$, show that α is a root of $P'(x) = 0$. **2**

- ii. The polynomial equation $x^5 - ax^2 + b = 0$ has a multiple root. ($a \neq 0, b \neq 0$). **3**

Show that $108a^5 = 3125b^3$.

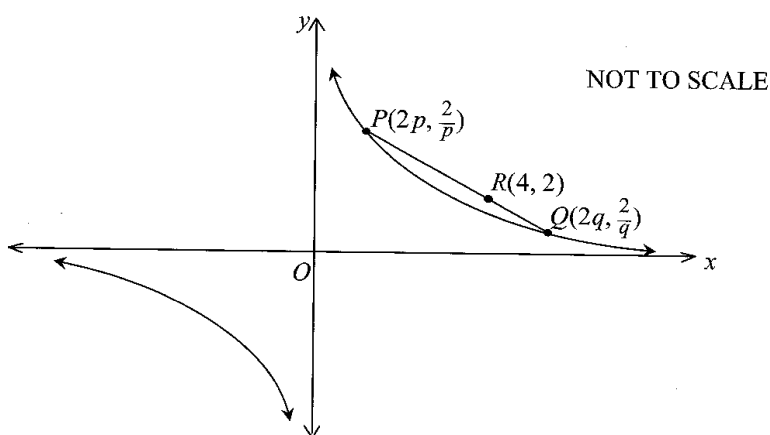
Question 5 (15 marks) Commence each question on a SEPARATE page

a. Find $\frac{dy}{dx}$ if $\sqrt{x} + \sqrt{y} = 1$. **2**

b. $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ are points on the rectangular hyperbola $xy = 4$.

M is the midpoint of the chord PQ .

P and Q move on the hyperbola so that the chord PQ always passes through the point $R(4, 2)$.



i. Show that the equation of the chord PQ is $x + p q y = 2(p + q)$. **2**

ii. Show that $p q = p + q - 2$. **1**

iii. Hence sketch the locus of M , as P and Q move on the curve $xy = 4$. **4**

c. i. If $z = \cos \theta + i \sin \theta$, explain why $z^n + z^{-n} = 2 \cos n\theta$ and $z^n - z^{-n} = 2i \sin n\theta$ for positive integers n . **2**

ii. By considering the binomial expansions of $(z + z^{-1})^3$ and $(z - z^{-1})^3$, show that $4(\cos^3 \theta + \sin^3 \theta) = (\cos 3\theta - \sin 3\theta) + 3(\cos \theta + \sin \theta)$. **2**

iii. Hence evaluate $\cos^3 \frac{\pi}{12} + \sin^3 \frac{\pi}{12}$. **2**

Question 6 (15 marks) Commence each question on a SEPARATE page

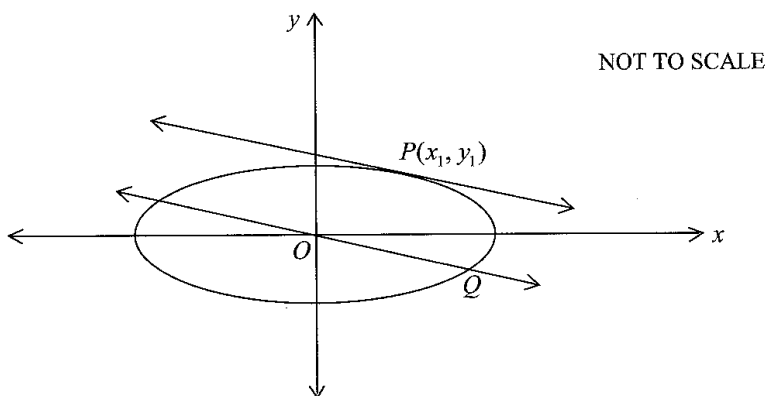
a. i. Verify that $\alpha = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ is a root of $z^5 + z - 1 = 0$. **2**

ii. Find the monic cubic equation with real coefficients whose roots are also roots of $z^5 + z - 1 = 0$ but do not include α . **3**

b. $P(x_1, y_1)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre O .

A line drawn from O , parallel to the tangent to the ellipse at P , meets the ellipse at Q .

The equation of the tangent to the ellipse at P is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.



i. Show that the equation of the line OQ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 0$. **2**

ii. Show that the coordinates of Q , in terms of x_1 and y_1 are $\left(\frac{ay_1}{b}, -\frac{bx_1}{a}\right)$. **3**

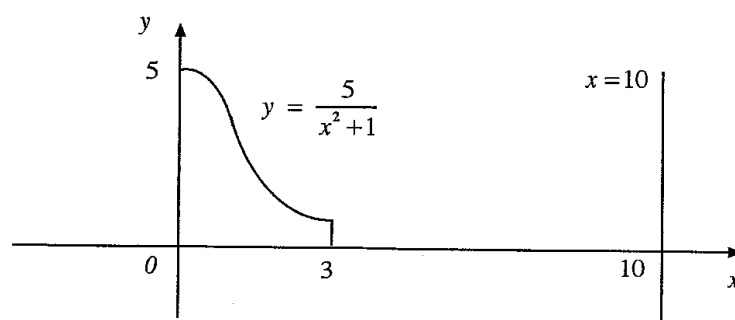
iii. Show that the distance between the tangent at P and the line OQ is $\frac{ab}{OQ}$. **4**

iv. Hence prove the area of the triangle OPQ is independent of the position P . **1**

Question 7 (15 marks) Commence each question on a SEPARATE page

- a. A particle of mass m is moving vertically in a resisting medium in which the resistance to motion has magnitude $\frac{1}{10}mv^2$ when the particle has speed $v \text{ ms}^{-1}$. The acceleration due to gravity is 10 ms^{-2} .
- i. If the particle projected vertically upwards from rest with speed $U \text{ ms}^{-1}$. **1**
 Show that during its upwards motion, its acceleration $a \text{ ms}^{-2}$ is given
 by $a = -\frac{1}{10}(100 + v^2)$
- ii. Hence show that its maximum height, H metres, is given **3**
 by $H = 5 \ln \left(\frac{U^2 + 100}{100} \right)$.
- iii. The particle falls vertically from rest. Show that during its downward **1**
 motion its acceleration $a \text{ ms}^{-2}$ is given by $a = \frac{1}{10}(100 - v^2)$.
- iv. Hence show that it returns to its point of projection with speed $V \text{ ms}^{-1}$ **4**
 given by $V = \frac{10U}{\sqrt{U^2 + 100}}$.

b.

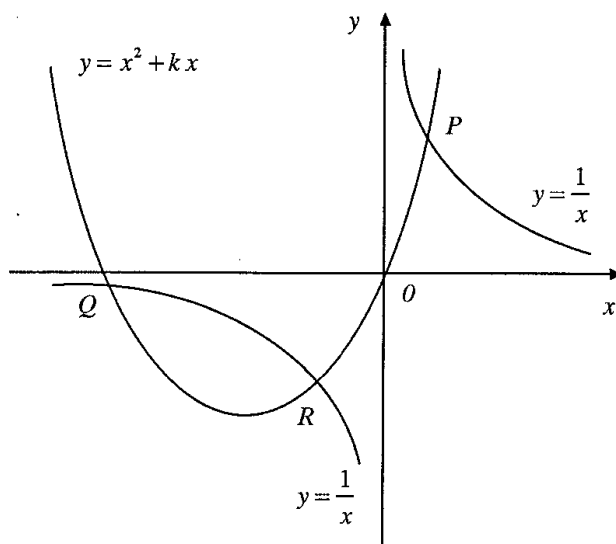


A circular flange is formed by rotating the region bounded by the curve $y = \frac{5}{x^2+1}$, the x axis and the lines $x = 0$ and $x = 3$, through one complete revolution about the line $x = 10$ (where all measurements are in cm).

- i. Use the method of cylindrical shells to show that the volume $V \text{ cm}^3$ of the flange is given by $V = \int_0^3 \frac{\pi(100-10x)}{x^2+1} dx$. **3**
- ii. Hence find the volume of the flange correct to the nearest cm^3 . **3**

Question 8 (15 marks) Commence each question on a SEPARATE page

a.



The curves $y = x^2 + kx$, where $k > 0$, and $y = \frac{1}{x}$ intersect at the points P , Q and R where $x = \alpha$, $x = \beta$ and $x = \gamma$.

- i. Show that α , β and γ are the roots of the equation $x^3 + kx^2 - 1 = 0$. **1**
- ii. Find the monic cubic equation with coefficients in terms of k whose roots are $-\alpha$, $-\beta$ and $-\gamma$. **1**
- iii. Find the monic cubic equation with coefficients in terms of k whose roots are $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$. **1**
- iv. Show that the gradient of PQ , QR and RP have a sum of k and a product of -1 . **2**

- b. A light inextensible string OP is fixed at one end O . A particle P is attached to the other end and moves uniformly in a horizontal circle whose centre is vertically below and at a distance h centimetres from O .
- i. By resolving the vertical and horizontal forces, show that $\omega = \sqrt{\frac{g}{h}}$ **2**
- ii. Hence, show that the period of the motion is given by $2\pi \frac{\sqrt{h}}{g}$. **1**
- iii. If the number of revolutions per second of the rotating particle is decreased from 2 rev/s to 1 rev/s, find the distance by which the level of the circle is lowered. **2**
- c. i. By using de Moivre's theorem and $(\cos \theta + i \sin \theta)^3$, or otherwise, show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ and $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ **2**
- ii. Hence, show $\tan 3\theta = \frac{t(3-t^2)}{1-3t^2}$, where $t = \tan \theta$. **3**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Question 1

$$\begin{aligned} \text{a. } \int_0^{\pi/4} \sec^2 x \tan^2 x \, dx \\ = \frac{1}{3} \tan^3 x \Big|_0^{\pi/4} \text{ by inspection} \\ = \frac{1}{3} [\tan^3 \pi/4 - \tan^3 0] \\ = \frac{1}{3} \end{aligned}$$

2

$$\begin{aligned} \text{b. } \int_0^3 \frac{5}{x^2 - 6x + 18} \, dx \\ = \int_0^3 \frac{5}{x^2 - 6x + 9 + 9} \, dx \\ = 5 \int_0^3 \frac{1}{9 + (x-3)^2} \, dx \\ = \frac{5}{3} \tan^{-1} \frac{x-3}{3} \Big|_0^3 \\ = \frac{5}{3} [\tan^{-1} 0 - \tan^{-1} -1] \\ = \frac{5}{3} [0 - -\pi/4] \\ = \frac{5\pi}{12} \end{aligned}$$

3

$$\text{c. i. } 4x - 6 = a(2x^2 + 3) + (bx + c)(x+1)$$

$$\text{Let } x = -1 \therefore -10 = 5a \therefore a = -2$$

$$\text{Let } x = 0 \therefore -6 = 3a + c$$

$$\therefore -6 = -6 + c \therefore c = 0$$

$$\text{Let } x = 2 \therefore 2 = 11a + 6b + 3c$$

$$\therefore 2 = -22 + 6b$$

$$6b = 24 \therefore b = 4$$

$$\therefore a = -2, b = 4, c = 0$$

3

$$\text{ii. } \int \frac{4x-6}{(x+1)(2x^2+3)} \, dx = \int \frac{-2}{x+1} + \frac{4x}{2x^2+3} \, dx$$

$$= -2 \ln(x+1) + \ln(2x^2+3) + c$$

$$= \ln \left[\frac{2x^2+3}{(x+1)^2} \right] + c$$

2

$$\begin{aligned} \text{d. } \int_0^1 \frac{2y}{1+2y} \, dy &= \int_0^1 \frac{1+2y-1}{1+2y} \, dy \\ &= \int_0^1 1 - \frac{1}{1+2y} \, dy \end{aligned}$$

$$= y - \frac{1}{2} \ln(1+2y) \Big|_0^1$$

$$= 1 - \frac{1}{2} \ln 3 - (0)$$

2

$$= 1 - \frac{1}{2} \ln 3$$

$$\text{e. } t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} [1 + \tan^2 \frac{x}{2}]$$

$$= \frac{1}{2} (1 + t^2)$$

$$\therefore \frac{dt}{dx} = \frac{1+t^2}{2} \therefore dx = \frac{2 \, dt}{1+t^2}$$

$$x = \pi/3 \therefore t = \frac{1}{\sqrt{3}} \text{ and } x = 0, t = 0$$

$$\int_0^{\frac{1}{\sqrt{3}}} \frac{2 \, dt}{13 + \frac{5(2t)}{1+t^2} + \frac{12(1-t^2)}{1+t^2}}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2 \, dt}{13 + 13t^2 + 10t + 12 - 12t^2}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2 \, dt}{t^2 + 10t + 25}$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{(t+5)^2}$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} (t+5)^{-2} \, dt$$

$$= -2(t+5)^{-1} \Big|_0^{\frac{1}{\sqrt{3}}}$$

$$= \left[\frac{-2}{t+5} \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= -2 \left[\frac{1}{\frac{1}{\sqrt{3}} + 5} - \frac{1}{5} \right]$$

$$= -2 \left[\frac{\sqrt{3}}{1+5\sqrt{3}} - \frac{1}{5} \right]$$

3

$$= -2 \left[\frac{5\sqrt{3} - 1 - 5\sqrt{3}}{5(1+5\sqrt{3})} \right]$$

$$= \frac{2}{5+25\sqrt{3}}$$

Question 2

$$\text{a. } (1-2i)^2 = 1 - 4i + 4i^2$$

$$= 1 - 4i - 4$$

$$= -3 - 4i$$

1

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b. $z^2 = 5 - 12i$ Let $z = x + iy$

$\therefore x^2 - y^2 + 2xyi = 5 - 12i$

$\therefore x^2 - y^2 = 5 \quad 2xy = -12$

$\therefore y = -\frac{6}{x}$

$\therefore x^2 - \frac{36}{x^2} = 5$

$x^4 - 5x^2 + 36 = 0$

$(x^2 - 9)(x^2 + 4) = 0$

$\therefore x = \pm 3 \quad \therefore y = \mp 2$

$\therefore z = 3 - 2i, -3 + 2i$ 2

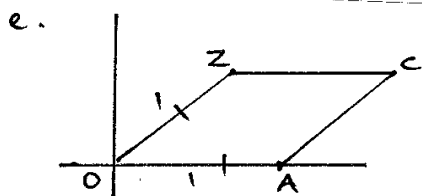
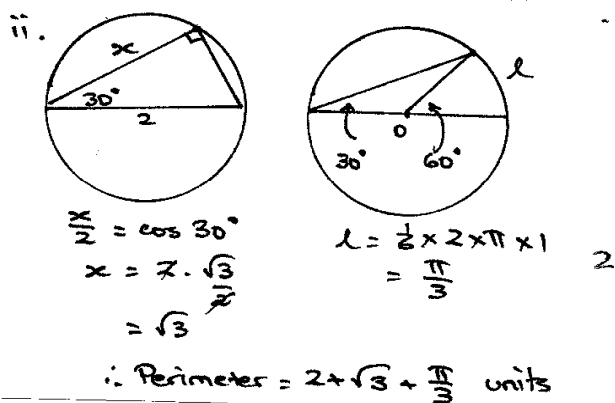
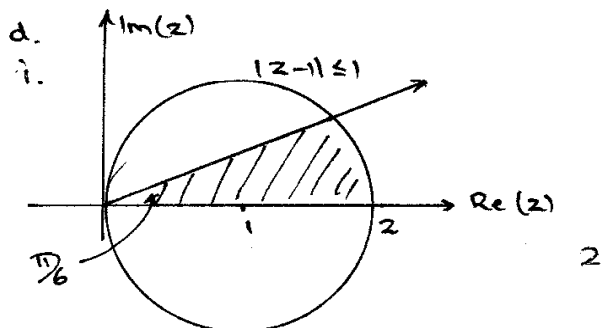
c. i. $1 + i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$ 1

ii. $(1 + i)^{12} = (2^{\frac{1}{2}})^{12} \operatorname{cis} 3\pi$

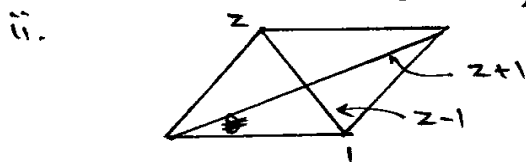
$= 2^6 \operatorname{cis} \pi$

$= 64 [\cos \pi + i \sin \pi]$

$= -64$ 2



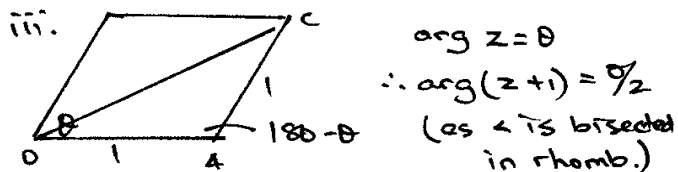
i. OACZ is parallelogram by addition of vectors. As $OA = OZ$, then OACZ is rhombus (adj sides equal) 1



Diags of rhombus are perpendicular

$\therefore \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$

$\therefore \frac{z-1}{z+1}$ is purely imaginary 2



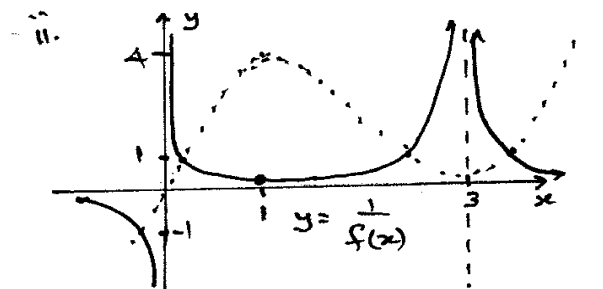
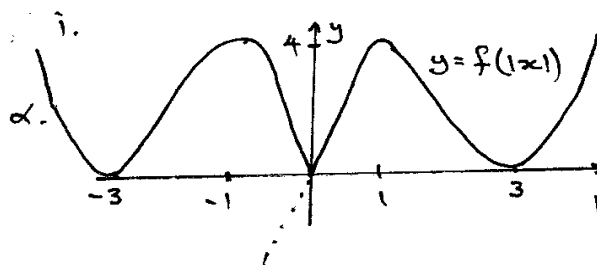
Now $|z+1| = \sqrt{1^2 + 1^2 - 2(1)(1) \cos(180-\theta)}$
 by cosine rule

$\therefore |z+1| = \sqrt{2+2 \cos \theta}$

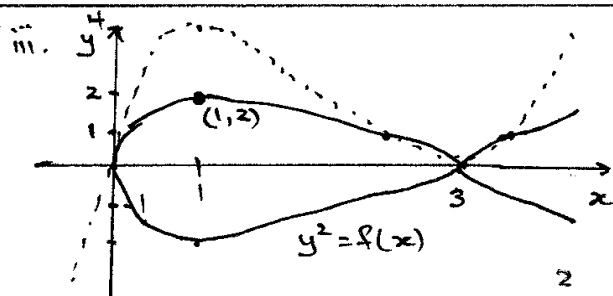
$\therefore \text{modulus} = \sqrt{2+2 \cos \theta}$

$\arg = \frac{\theta}{2}$ 2

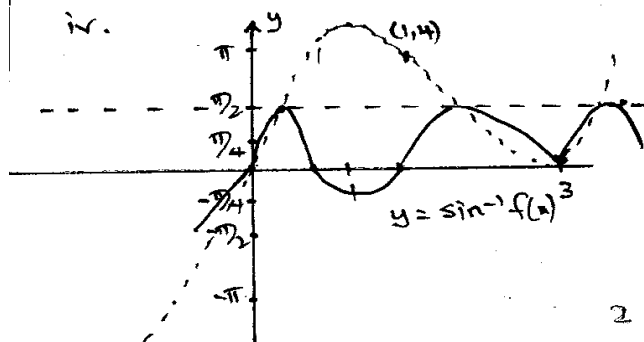
Question 3



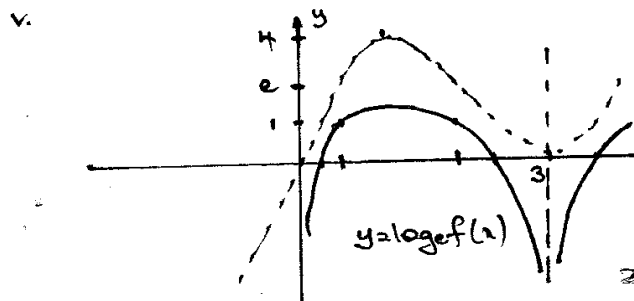
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2



2

[Note: $\pi/2 = 1.57$, $\pi = 3.14$, $4 > \pi$]

2

b. $x^2 - y^2 + xy + 5 = 0$

i. $2x - 2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$

$$(2y - x) \frac{dy}{dx} = 2x + y$$

$$\therefore \frac{dy}{dx} = \frac{2x + y}{2y - x}$$

2

ii. $y = x \therefore \text{grad} = 1$

$$\frac{2x + y}{2y - x} = 1$$

$$2x + y = 2y - x$$

$$y = 3x \quad \text{--- ①}$$

Also, $x^2 - y^2 + xy + 5 = 0$ --- ②

Subs ① in ②

$$x^2 - 9x^2 + x(3x) + 5 = 0$$

$$x^2 - 9x^2 + 3x^2 + 5 = 0$$

$$5x^2 = 5 \therefore x = \pm 1$$

Subs in ① $y = \pm 3$

$$\therefore (1, 3) \text{ and } (-1, -3)$$

2

c. Solve simultaneously,

$$y = -3x \text{ and } y = x + \frac{1}{x-1}$$

$$\therefore x + \frac{1}{x-1} = -3x$$

$$\therefore x(x-1) + 1 = -3x(x-1)$$

$$x^2 - x + 1 = -3x^2 + 3x$$

$$4x^2 - 4x + 1 = 0$$

$$(2x - 1)^2 = 0$$

$$\therefore x = \frac{1}{2} \text{ is only soln}$$

$$\therefore \left(\frac{1}{2}, -\frac{3}{2}\right) \text{ is the point of}$$

$$\text{contact} \therefore \text{tangent}$$

3

Question 4

a. i. $I_n = \int_0^{\pi/2} \cos^n x \, dx$

$$= \int_0^{\pi/2} \cos^{n-1} x \cdot \cos x \, dx$$

$$= \cos^{n-1} x \cdot \sin x \Big|_0^{\pi/2}$$

$$- \int_0^{\pi/2} (n-1) \cos^{n-2} x \cdot (-\sin x) \cdot \sin x \, dx$$

$$= 0 + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin^2 x \, dx$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x \, dx - (n-1) \int_0^{\pi/2} \cos^n x \, dx$$

$$= (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2} \quad 3$$

ii. $\int_0^{\pi/2} \cos^4 x \, dx = I_4$

$$\therefore I_4 = \frac{3}{4} I_2$$

$$\text{and } I_2 = \frac{1}{2} I_0$$

Now, $I_0 = \int_0^{\pi/2} \cos^0 x \, dx$

$$= \int_0^{\pi/2} 1 \, dx$$

$$= x \Big|_0^{\pi/2} = \frac{\pi}{2}$$

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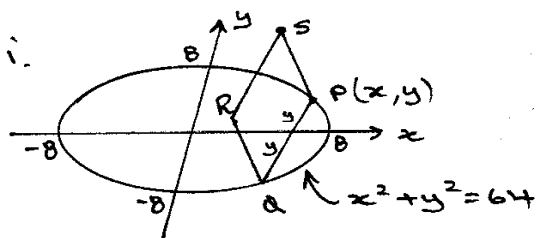
$$\therefore I_0 = \pi_2$$

$$\therefore I_2 = \frac{1}{2} (\pi_2) \\ = \pi_4$$

$$I_4 = \frac{3}{4} (\pi_4) \\ = 3\pi_{16}$$

3

b. i.



$$PQ = 2y$$

$$\therefore \text{Area PQRS} = 4y^2 \\ = 4[64 - x^2]$$

2

ii. Volume = Sum of areas

$$\therefore V = \int_{-8}^8 4(64 - x^2) dx$$

$$= 8 \int_0^8 (64 - x^2) dx$$

$$= 8 \left[64x - \frac{x^3}{3} \right]_0^8$$

$$= 8 \left[512 - \frac{512}{3} \right]$$

$$= 2730 \frac{2}{3} \text{ cm}^3$$

2

c. i. If α is double root,

$$P(x) = (x - \alpha)^2 Q(x)$$

$$P'(x) = 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x) \\ = (x - \alpha) [2Q(x) + (x - \alpha)Q'(x)]$$

$$\therefore P'(\alpha) = 0$$

2

ii. Let $P(x) = x^5 - ax^2 + b$

$$P'(x) = 5x^4 - 2ax = 0$$

$$x(5x^3 - 2a) = 0$$

$$\therefore x = 0, \quad x = \sqrt[3]{\frac{2a}{5}}$$

$$\text{But } P(0) \neq 0$$

$$\therefore x = \sqrt[3]{\frac{2a}{5}} \text{ is root}$$

Now subs:

$$P\left(\left(\frac{2a}{5}\right)^{\frac{1}{3}}\right) = \left(\frac{2a}{5}\right)^{\frac{5}{3}} - a\left(\frac{2a}{5}\right)^{\frac{2}{3}} - b = 0$$

$$a^{\frac{5}{3}} \left[\left(\frac{2}{5}\right)^{\frac{5}{3}} - \left(\frac{2}{5}\right)^{\frac{2}{3}} \right] = -b$$

$$a^{\frac{5}{3}} \cdot \left(\frac{2}{5}\right)^{\frac{2}{3}} \left[\frac{2}{5} - 1 \right] = -b$$

cube both sides:

$$a^5 \cdot \left(\frac{2}{5}\right)^2 \left[-\frac{3}{5} \right]^3 = -b^3$$

$$\frac{-108a^5}{3125} = -b^3$$

$$\therefore 108a^5 = 3125b^3$$

3

Question 5

$$a. \quad x^{\frac{1}{2}} + y^{\frac{1}{2}} = 1$$

$$\therefore \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} \frac{dy}{dx} = 0$$

$$\frac{1}{\sqrt{y}} \frac{dy}{dx} = -\frac{1}{\sqrt{x}}$$

$$\frac{dy}{dx} = -\sqrt{\frac{y}{x}}$$

2

$$b. \text{ grad } PQ = \frac{\frac{2}{p} - \frac{2}{q}}{\frac{2p-2q}{2p \cdot 2q}}$$

i.

$$= \frac{2q^{-1} - 2p^{-1}}{p^2} \times \frac{1}{2p-2q} \\ = \frac{-1}{p^2 q}$$

$$\therefore y - \frac{2}{p} = -\frac{1}{p^2 q} (x - 2p)$$

$$p^2 y - 2p = -x + 2p$$

$$\therefore x + p^2 y = 2(p + q) \quad \text{--- (1)}$$

2

ii. subs R into (1)

$$\therefore 4 + 2p^2 q = 2p + 2q$$

$$p^2 q = p + q - 2 \quad \text{--- (2)}$$

1

iii. Midpoint: M

$$\left(\frac{2p+2q}{2}, \frac{\frac{2}{p} + \frac{2}{q}}{2} \right)$$

$$= \left(p+q, \frac{2q+2p}{p^2 q} \times \frac{1}{2} \right)$$

$$= \left(p+q, \frac{p+q}{p^2 q} \right)$$

$$\therefore \text{Let } x = p+q \quad y = \frac{p+q}{p^2 q}$$

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From (3) $pq = p + q - 2$

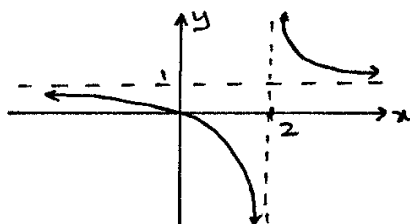
$\therefore pq = x - 2$

Now, from $y = \frac{p+q}{pq}$

$\therefore y = \frac{x}{x-2}$

$= \frac{x-2+2}{x-2}$

$= 1 + \frac{2}{x-2}$



c. i. $z = \cos \theta + i \sin \theta$

By de Moivre's theorem:

$z^n = \cos n\theta + i \sin n\theta$ — (1)

$z^{-n} = \cos -n\theta + i \sin -n\theta$
 $= \cos n\theta - i \sin n\theta$ — (2)

(1) + (2) $z^n + z^{-n} = 2 \cos n\theta$

(1) - (2) $z^n - z^{-n} = 2i \sin n\theta$

ii. $(z + z^{-1})^3 = z^3 + 3z + 3z^{-1} + z^{-3}$
 $= (z^3 + z^{-3}) + 3(z + z^{-1})$

$(2 \cos \theta)^3 = 2 \cos 3\theta + 6 \cos \theta$

$8 \cos^3 \theta = 2 \cos 3\theta + 6 \cos \theta$

$\therefore 4 \cos^3 \theta = \cos 3\theta + 3 \cos \theta$ — (3)

Similarly,

$(z - z^{-1})^3 = z^3 - 3z + 3z^{-1} - z^{-3}$

$= (z^3 - z^{-3}) - 3(z - z^{-1})$

$\therefore (2i \sin \theta)^3 = 2i \sin 3\theta - 6i \sin \theta$

$8i^3 \sin^3 \theta = 2i \sin 3\theta - 6i \sin \theta$

$-4i \sin^3 \theta = i \sin 3\theta - 6i \sin \theta$

$\therefore -4 \sin^3 \theta = \sin 3\theta - 6 \sin \theta$ — (4)

(3) - (4) $4(\cos^3 \theta + \sin^3 \theta)$

$= (\cos 3\theta + \sin 3\theta) + 3(\cos \theta + \sin \theta)$

iii. From ii,

$\cos^3 \pi_{12} + \sin^3 \pi_{12}$

$= \frac{1}{4} [\cos \pi_4 - \sin \pi_4 + 3(\cos \pi_{12} + \sin \pi_{12})]$

$= \frac{3}{4} [\cos \pi_{12} + \sin \pi_{12}]$

$= \frac{3}{4} \cdot \sqrt{2} [\frac{1}{\sqrt{2}} \cos \pi_{12} + \frac{1}{\sqrt{2}} \sin \pi_{12}]$

$= \frac{3\sqrt{2}}{4} [\sin \pi_4 \cos \pi_{12} + \cos \pi_4 \sin \pi_{12}]$

$= \frac{3\sqrt{2}}{4} [\sin(\pi_4 + \pi_{12})]$

$= \frac{3\sqrt{2}}{4} [\sin \pi_3]$

$= \frac{3\sqrt{2}}{4} \cdot \frac{\sqrt{3}}{2}$

$= \frac{3\sqrt{6}}{8}$

Question 6

a. i. $\alpha = \cos \pi_3 + i \sin \pi_3$

$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$ — (1)

Now, $\alpha^5 = \cos 5\pi_3 + i \sin 5\pi_3$

$= \cos \pi_3 - i \sin \pi_3$

$= \frac{1}{2} - \frac{\sqrt{3}}{2}i$ — (2)

$\therefore p(z) = z^5 + z - 1$

$= \alpha^5 + \alpha - 1$

$= \frac{1}{2} - \frac{\sqrt{3}}{2}i + \frac{1}{2} + \frac{\sqrt{3}}{2}i - 1 = 0$

$\therefore \alpha$ is a root

ii. Roots in conjugate pairs:

$(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$ and $(\frac{1}{2} - \frac{\sqrt{3}}{2}i)$

$\therefore$ Quadratic is

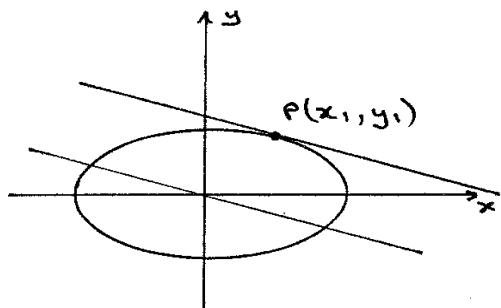
$z^2 - (\text{sum of roots})z + (\text{prod of roots}) = 0$

$z^2 - z + 1$

$$\begin{array}{r} z^2 - z + 1 \overline{) 2^5 + 0z^4 + 0z^3 + 0z^2 + 2z - 1} \\ \underline{2^5 - z^4 + z^3} \\ 2^4 - z^3 + 0z^2 \\ \underline{2^4 - z^3 + z^2} \\ -z^2 + z - 1 \\ \underline{-z^2 + z - 1} \\ 0 \end{array}$$

$\therefore z^3 + z^2 - 1$

b.

i. Tangent at P: $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ Now, if OQ is parallel, $\therefore$ of the form

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = c$$

Now, OQ passes through (0,0)

$$\therefore 0+0=c \quad \therefore c=0$$

$$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 0 \quad \text{--- (1)} \quad 2$$

$$\text{ii. From (1)} \quad \frac{yy_1}{b^2} = -\frac{xx_1}{a^2}$$

$$y = -\frac{b^2 x_1 x}{a^2 y_1} \quad \text{--- (2)}$$

$$\text{and } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (3)}$$

Subs (2) in (3)

$$\frac{x^2}{a^2} + \frac{b^2 x_1^2 x^2}{a^4 y_1^2} = 1$$

$$x^2 \left[\frac{1}{a^2} + \frac{b^2 x_1^2}{a^4 y_1^2} \right] = 1$$

$$x^2 \left[\frac{a^2 y_1^2 + b^2 x_1^2}{a^4 y_1^2} \right] = 1 \quad \text{--- (4)}$$

$$\text{Now, from (3)} \quad b^2 x^2 + a^2 y^2 = a^2 b^2$$

and subs in (x_1, y_1) :

$$b^2 x_1^2 + a^2 y_1^2 = a^2 b^2$$

Now subs in (4)

$$x^2 \left[\frac{a^2 b^2}{a^4 y_1^2} \right] = 1$$

$$x^2 = \frac{a^4 y_1^2}{a^2 b^2}$$

$$\therefore x = \frac{a^2 y_1}{ab}$$

$$\therefore x = \frac{a y_1}{b}$$

$$\text{Subs in (2)} \quad y = \frac{-b^2 x_1}{a^2 y_1} \cdot \frac{a y_1}{b}$$

$$= -\frac{b x_1}{a}$$

$$\therefore \left(\frac{a y_1}{b}, -\frac{b x_1}{a} \right) \quad 3$$

iii. Use perpendicular dist. formula:

$$d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$= \frac{\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} + 0}{\sqrt{\frac{x_1^2}{a^4} + \frac{y_1^2}{b^4}}} \quad \leftarrow \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

$$= 1 \div \sqrt{\frac{b^4 x_1^2 + a^4 y_1^2}{a^4 b^4}}$$

$$= \frac{a^2 b^2}{\sqrt{x_1^2 b^4 + y_1^2 a^4}} \quad \text{--- (A)}$$

Now, length OQ:

$$\sqrt{\left(\frac{a y_1}{b} - 0 \right)^2 + \left(-\frac{b x_1}{a} - 0 \right)^2}$$

$$= \sqrt{\frac{a^2 y_1^2}{b^2} + \frac{b^2 x_1^2}{a^2}}$$

$$= \sqrt{\frac{a^4 y_1^2 + b^4 x_1^2}{a^2 b^2}}$$

$$= \frac{\sqrt{a^4 y_1^2 + b^4 x_1^2}}{ab}$$

$$\text{Now (A)} = \frac{ab}{\text{(B)}}$$

$$\therefore \text{distance is } \frac{ab}{OQ} \quad 4$$

$$\text{iv. Area } \triangle OPQ = \frac{1}{2} \times d \times OQ$$

$$= \frac{1}{2} \times \frac{ab}{OQ} \times OQ$$

$$= \frac{1}{2} ab \text{ which is independent of } P. \text{ (ie of } x_1, y_1)$$

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Question 7:



$$m\ddot{x} = -10m - \frac{1}{10}mv^2$$

$$= -\frac{1}{10}(100 + v^2)$$

$$ii. v \frac{dv}{dx} = -\frac{1}{10}(100 + v^2)$$

$$\frac{dv}{dx} = -\frac{(100 + v^2)}{10v}$$

$$\frac{dy}{dv} = \frac{-10v}{100 + v^2}$$

$$x = -5 \ln(100 + v^2) + c$$

$$x=0, v=U$$

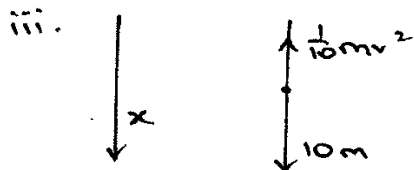
$$\therefore 0 = -5 \ln(100 + U^2) + c$$

$$c = 5 \ln(100 + U^2)$$

$$\therefore x = 5 \ln \left(\frac{100 + U^2}{100 + v^2} \right)$$

Max height when $v=0$

$$\therefore x = 5 \ln \left(\frac{100 + U^2}{100} \right) \quad \text{--- ①} \quad 3$$



$$m\ddot{x} = 10m - \frac{1}{10}mv^2$$

$$\ddot{x} = \frac{1}{10}(100 - v^2)$$

$$iv. v \frac{dv}{dx} = \frac{1}{10}(100 - v^2)$$

$$\frac{dv}{dx} = \frac{100 - v^2}{10v}$$

$$\frac{dy}{dv} = \frac{10v}{100 - v^2}$$

$$x = -5 \ln(100 - v^2) + c$$

Let particle fall from $x=0$ ***

$$\therefore x=0, v=0$$

$$\therefore 0 = -5 \ln 100 + c$$

$$\therefore c = 5 \ln 100$$

$$\therefore x = 5 \ln \left(\frac{100}{100 - v^2} \right)$$

Let $v=V$

$$x = 5 \ln \frac{100}{100 - V^2} \quad \text{--- ②}$$

Now ① = ②

$$\therefore 5 \ln \left(\frac{100 + U^2}{100} \right) = 5 \ln \left(\frac{100}{100 - v^2} \right)$$

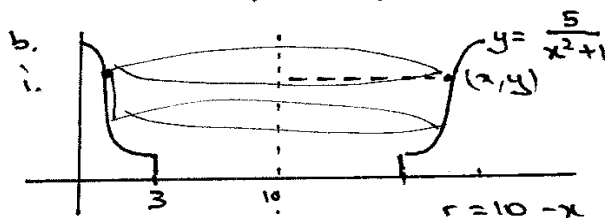
$$\therefore \frac{100 + U^2}{100} = \frac{100}{100 - v^2}$$

$$100 - v^2 = \frac{10000}{100 + U^2}$$

$$v^2 = 100 - \frac{10000}{100 + U^2}$$

$$= \frac{10000 + 100U^2 - 10000}{U^2 + 100}$$

$$v = \frac{10U}{\sqrt{U^2 + 100}} \quad 4$$



$$V = \int 2\pi r h \, dx$$

$$= 2\pi \int_0^3 (10 - x) \cdot y \, dx$$

$$= 2\pi \int_0^3 (10 - x) \cdot \frac{5}{x^2 + 1} \, dx$$

$$= \int_0^3 \frac{\pi(100 - 10x)}{x^2 + 1} \, dx \quad 3$$

$$ii. V = 100\pi \int_0^3 \frac{1}{x^2 + 1} \, dx - 5\pi \int_0^3 \frac{2x}{x^2 + 1} \, dx$$

$$= 100\pi \tan^{-1} x \Big|_0^3 - 5\pi \ln(x^2 + 1) \Big|_0^3$$

$$= 100\pi \tan^{-1} 3 - 5\pi \ln 10$$

$$= 356 \text{ cm}^3 \quad 3$$

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Question 8:

a i. $y = x^2 + kx$ — ①

$y = \frac{1}{x}$ — ②

① = ② $x^2 + kx = \frac{1}{x}$

$\therefore x^3 + kx^2 = 1$

$\therefore x^3 + kx^2 - 1 = 0$

 $\therefore \alpha, \beta, \gamma$ are roots of equation.ii. Subs $-x$ into $x^3 + kx^2 - 1 = 0$

$\therefore (-x)^3 + k(-x)^2 - 1 = 0$

$-x^3 + kx^2 - 1 = 0$

$\therefore x^3 - kx^2 + 1 = 0$

iii. Subs $\frac{1}{x}$ into $x^3 + kx^2 - 1 = 0$

$\therefore \left(\frac{1}{x}\right)^3 + k\left(\frac{1}{x}\right)^2 - 1 = 0$

$\frac{1}{x^3} + \frac{k}{x^2} - 1 = 0$

$1 + kx - x^3 = 0$

$\therefore x^3 - kx - 1 = 0$

iv. $P(\alpha, \frac{1}{\alpha}), Q(\beta, \frac{1}{\beta}), R(\gamma, \frac{1}{\gamma})$

grad PQ: $\frac{\frac{1}{\alpha} - \frac{1}{\beta}}{\alpha - \beta} = \frac{\beta - \alpha}{\alpha\beta} \times \frac{1}{\alpha - \beta}$
 $= \frac{-1}{\alpha\beta}$

Similarly, other gradients

$-\frac{1}{\alpha\gamma}$ and $-\frac{1}{\beta\gamma}$

$\therefore \text{Sum: } -\frac{1}{\alpha\beta} - \frac{1}{\alpha\gamma} - \frac{1}{\beta\gamma}$

$= \frac{-\gamma - \beta - \alpha}{\alpha\beta\gamma}$

$= \frac{-(\alpha + \beta + \gamma)}{\alpha\beta\gamma}$

$= \frac{-(-k)}{1}$

$= k$

Product = $-\frac{1}{\alpha\beta} \cdot -\frac{1}{\alpha\gamma} \cdot -\frac{1}{\beta\gamma}$

$= \frac{-1}{\alpha^2\beta^2\gamma^2}$

$= \frac{-1}{(\alpha\beta\gamma)^2}$

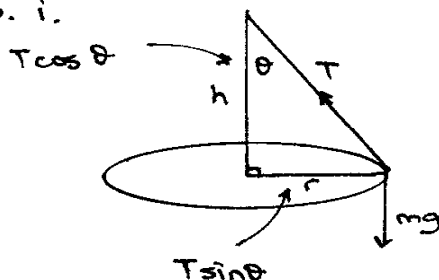
$= \frac{-1}{1}$

$= -1$

$\therefore \text{sum} = k, \text{product} = -1$

2

b. i.



Vertically: $\Sigma = 0$

$\therefore T \cos \theta - mg = 0$

$\therefore T \cos \theta = mg$ — ①

Horizontally: $\Sigma = mr\omega^2$

$\therefore T \sin \theta = mr\omega^2$ — ②

② $\div$ ① $\tan \theta = \frac{r\omega^2}{g}$

$\therefore \tan \theta = \frac{r\omega^2}{g}$

But $\tan \theta = \frac{r}{h}$

$\therefore \frac{r\omega^2}{g} = \frac{r}{h}$

$\omega^2 = \frac{g}{h}$

2

$\therefore \omega = \sqrt{\frac{g}{h}}$

ii. Period = $\frac{2\pi}{\omega}$

$= 2\pi \div \sqrt{\frac{g}{h}}$

$= 2\pi \sqrt{\frac{h}{g}}$

1

iii. $\omega = \text{angular velocity}$

$\therefore 2 \text{ rev/sec} = 2 \times 2\pi = 4\pi$

and $1 \text{ rev/sec} = 2\pi$

From i. $\omega^2 = \frac{g}{h}$

$\therefore h = \frac{g}{\omega^2}$

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$$w = 4\pi, \quad h = \frac{9}{16\pi^2}$$

$$w = 2\pi, \quad h = \frac{9}{4\pi^2}$$

$$\begin{aligned} \therefore \text{Difference} &= \frac{9}{16\pi^2} - \frac{9}{4\pi^2} \\ &= \frac{9}{16\pi^2} - \frac{4 \cdot 9}{16\pi^2} \\ &= \frac{-3 \cdot 9}{16\pi^2} \end{aligned}$$

$$\therefore \text{lowered by } \frac{3 \cdot 9}{16\pi^2} \text{ cm} \quad 2$$

$$c. i. (\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta \quad \text{--- ①}$$

$$\begin{aligned} \text{or } (\cos \theta + i \sin \theta)^3 &= \cos^3 \theta + 3 \cos^2 \theta \cdot i \sin \theta \\ &\quad + 3 \cos \theta \cdot i^2 \sin^2 \theta + i^3 \sin^3 \theta \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta \\ &\quad - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \quad \text{--- ②} \end{aligned}$$

As ① = ② and equating real + imag.

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta + 3 \cos^3 \theta - 3 \cos \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta \end{aligned}$$

$$\begin{aligned} \sin 3\theta &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta \\ &= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta \\ &= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta \quad 2 \end{aligned}$$

$$\begin{aligned} ii. \tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{3 \sin \theta - 4 \sin^3 \theta}{4 \cos^3 \theta - 3 \cos \theta} \\ &= \frac{\sin \theta (3 - 4 \sin^2 \theta)}{\cos \theta (4 \cos^2 \theta - 3)} \end{aligned}$$

Divide top + bottom by $\cos^2 \theta$

$$= \tan \theta \left[\frac{\frac{3}{\cos^2 \theta} - 4 \tan^2 \theta}{4 - \frac{3}{\cos^2 \theta}} \right]$$

$$\begin{aligned} &= \tan \theta \left[\frac{3 \sec^2 \theta - 4 \tan^2 \theta}{4 - 3 \sec^2 \theta} \right] \\ &= \tan \theta \left[\frac{3(1 + \tan^2 \theta) - 4 \tan^2 \theta}{4 - 3(1 + \tan^2 \theta)} \right] \\ &= \tan \theta \left[\frac{3 - \tan^2 \theta}{1 - 3 \tan^2 \theta} \right] \\ &= t \left[\frac{3 - t^2}{1 - 3t^2} \right] \\ &= \frac{t(3 - t^2)}{1 - 3t^2} \quad 3 \end{aligned}$$